Fertility, Housing, and Population Dynamics

From a stationary closure to a dated transition

Tommaso De Santo

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The household mechanism stays; the population closure changes

The model shown to Raquel

- Sequential birth attempts and declining fecundity
- Children ever born and children at home tracked separately
- Rent versus own, discrete house sizes, saving, income risk, and bequests
- Housing costs affect fertility through space needs and tenure access

The missing aggregate object

- How current births generate future households
- How cohort sizes evolve away from stationarity
- How housing clears as population changes
- Where the observed economy sits on that path

Household choices $\longrightarrow$ births $\longrightarrow$ entrants $\longrightarrow$ housing demand $\longrightarrow$ prices $\longrightarrow$ choices

This is a new equilibrium closure, not a new household model.

A two-generation model isolates the feedback

Young households choose consumption, adult space, and children:

$$u(c, h, n) = c + \alpha \log h + v \log n,$$

$$c + ph + \left[q + a(p + \omega)\right] n = y.$$

Their demands are

$$n(p) = \frac{v}{q + a(p + \omega)}, \qquad h_Y(p) = \frac{\alpha}{p} + an(p).$$

Old households retain housing:

$$h_O(p) = \frac{\bar{h}_O}{p}.$$

Housing supply is

$$H^s(p) = \bar{H} p^{\varepsilon}.$$

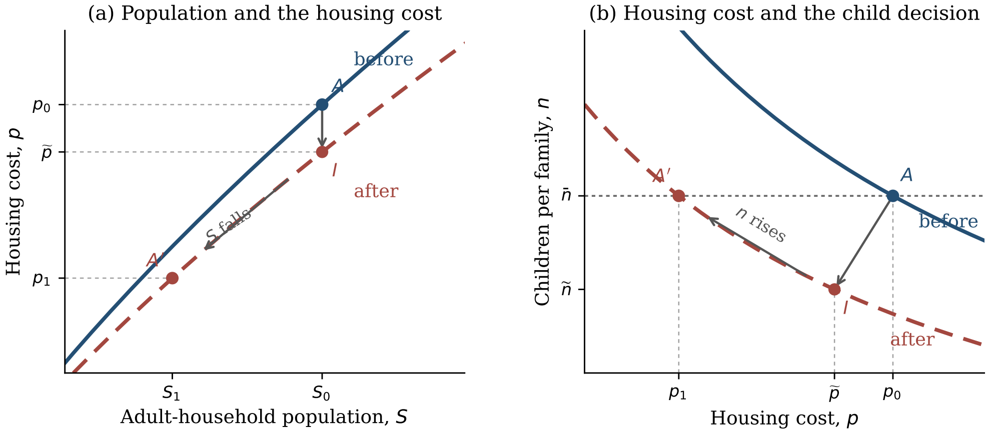
Demography evolves according to

$$Y_{t+1} = \frac{n(p_t)}{\bar{n}} Y_t, \qquad O_{t+1} = Y_t.$$

Economic content. A higher housing cost raises the cost of children; fewer children reduce future household formation; a smaller population eventually lowers housing costs.

Source: two-generation model in the current paper. The objects a, q, ω, and v summarize the full model's child-space requirement, other child costs, tenure frictions, and fertility preference.

The steady state combines replacement fertility and housing clearing



- The child decision pins down the housing cost consistent with replacement: $n(p^*) = \bar{n}$.
- Housing clearing then determines how many households can live at that cost.

Blue: initial value of children. Red: permanently lower value. The arrows compare impact and steady-state outcomes; they are not the literal transition path.

Below-replacement fertility need not lower population immediately

Let $m_t = n(p_t)/\bar{n}$ be the replacement ratio and $\mu_t = O_t/Y_t$ the old-to-young cohort ratio. Population can rise with $m_t < 1$ whenever

$$\mu_t < m_t < 1.$$

A numerical example

$$100 \text{ young} + 70 \text{ old} \quad \longrightarrow \quad 80 \text{ young} + 100 \text{ old}$$

$m_t = 0.8 < 1$, but total households rise from 170 to 180.

Housing costs can rise as well if the aging cohort retains more housing than is released by the shrinking young cohort:

$$(1-\mu_t)h_O(p_t) > (1-m_t)h_Y(p_t).$$

U.S. interpretation: inherited cohorts can keep population and housing costs rising even after fertility falls below replacement.

The full model needs a dated distribution law, not a redesign

Keep the existing household block

- Sequential fertility and child maturation
- Tenure, house size, saving, and moving costs
- Earnings risk, retirement, survival, and bequests

Add three aggregate objects

1. A calendar-time distribution $g_t(j, x)$ over age and household states
2. A birth-vintage queue carrying births to entry twenty years later
3. Housing-market clearing at every date

$$B_t = \frac{b_t^{\text{adjusted}}}{2.1}, \qquad e_{t+1} = M + \rho B_{t-4}, \qquad g_{t+1} = \mathcal{Q}_t g_t + \text{entrants}_{t+1}.$$

$$H^s(P_t) = \int h_t(j, x)\, dg_t(j, x).$$

The factor 2.1 **is explicit replacement accounting:** adjusted child units per future household, combining sex composition, survival to entry age, and household formation. There is no hidden child-death state.

We calibrate 2007–2023 as a transition, not as a steady state

Old benchmark $\longrightarrow$	**2007 state** $\longrightarrow$	**2007–2023** $\longrightarrow$	**After 2023**
Completed fertility normalized to 2.1	Reweight only the age margin to ACS 2007	Estimate a linear decline in the fertility preference	Hold the 2023 preference fixed and continue

Imposed inputs

- Census household totals
- ACS head-age shares
- 2007 start and linear path shape

Calibration

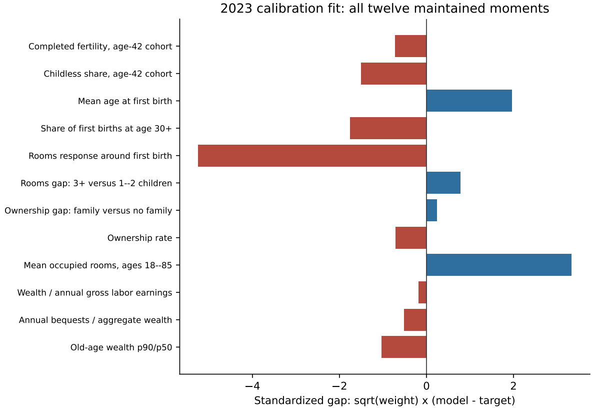
- Ten free structural parameters
- Twelve maintained moments
- Objective evaluated mainly at 2023

Historical holdouts

- Period fertility and timing
- Ownership and rooms
- House prices and rents

Thus 2007 is a dated initial state, not literally the old steady state, and the demographic bridge is not counted as model fit.

The 2023 fit has two important housing-size misses



Calibration facts

Loss: 50.742. Ten parameters and twelve moments.

Two exact independent repeats.

Main misses

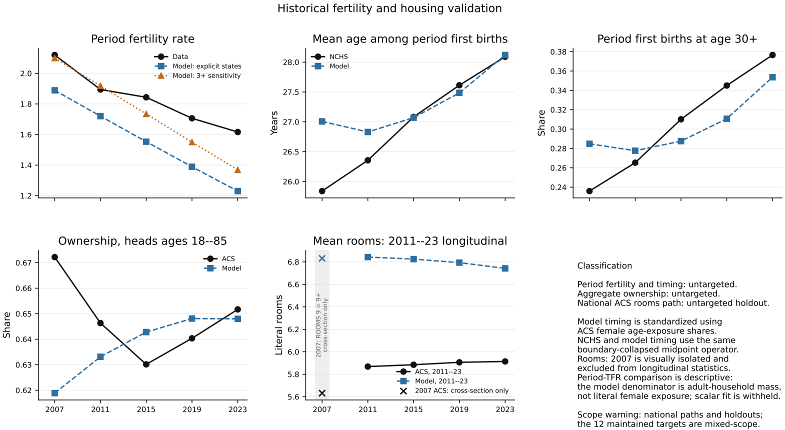
- First-birth rooms: 0.273 versus 0.720
- Mean rooms: 6.742 versus 5.780

They contribute 38.57 of the total loss.

The PSID rooms target is provisional because its prepath is not flat.

Source: final dated-transition calibration, selected candidate 7; target system `e5_fullhistory_roomsfix_h1_20260817`.

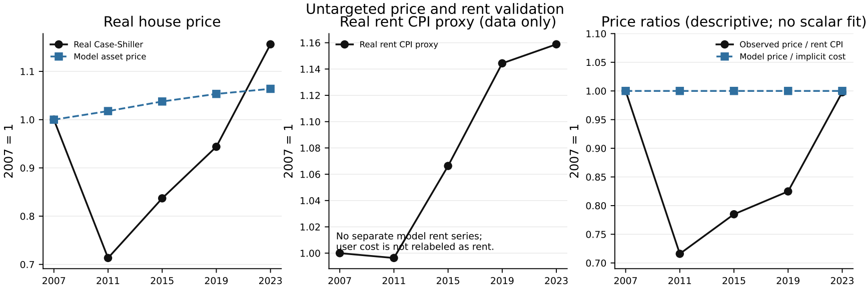
Historical validation is mixed rather than hidden



Reading the figure: timing is close by 2023; the model misses the ownership cycle and rooms trend, and its fertility decline is too steep.

National historical series are holdouts. The maintained 2023 targets use mixed samples and vintages, so this is not a unified national time-series fit.

The model does not reproduce the house-price cycle

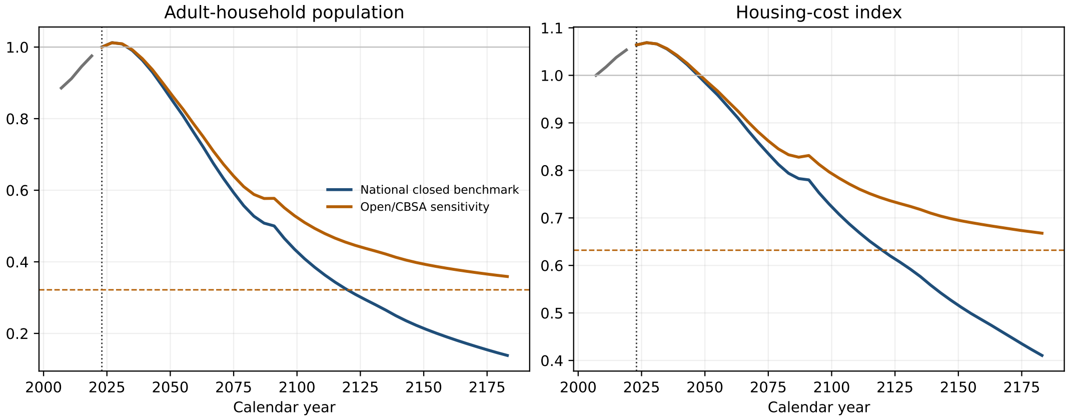


- The model cannot reproduce the 2007–2011 house-price collapse and later rebound.
- With temporary-equilibrium expectations and a fixed user-cost relation, it has no independent price-to-rent dynamics.

The exercise should therefore be read as a demographic-housing transition, not as a complete account of the post-2007 housing cycle.

Observed house prices: real Case-Shiller. The rent CPI is shown only as a descriptive proxy, not as the model's user cost.

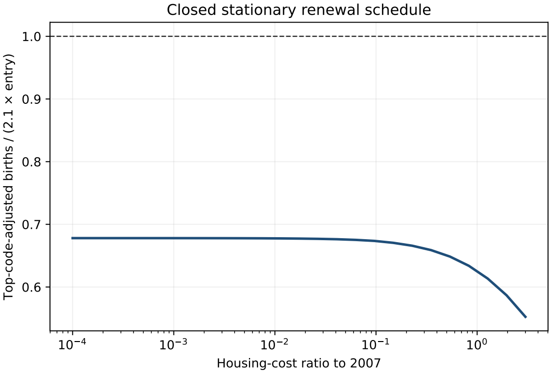
Demographic momentum delays the post-2023 decline



Both paths peak slightly in 2027. By 2183, population is 0.138 of 2023 in the closed benchmark and 0.359 in the open sensitivity; asset prices are 0.386 and 0.628 of 2023.

The open case fixes the old-state outside flow M and retention rate ρ. The value 0.169 normalizes the old state; it is not a constant entrant share along the path.

No positive closed steady state at the fitted 2023 preference



A closed steady state requires

$$\frac{B(P)}{E(P)} = 1.$$

At the estimated 2023 preference,

$$0.552 \leq B(P)/E(P) \leq 0.678$$

over the declared price grid. There is no crossing.

Even almost free housing does not restore replacement in the fitted model.

Therefore: the blue path is a finite-horizon closed benchmark, not a verified transition between two positive steady states.

Source: 25-point grid, $P/P_0 \in [10^{-4}, 3]$; not a global theorem.

The honest advisor summary

What the update adds

The paper keeps the household block and adds population renewal. It treats 2007–2023 as a dated transition, calibrates at 2023, and follows inherited cohorts forward.

- **Theory:** lower fertility eventually reduces population and housing costs, but inherited cohorts can keep population and prices rising initially.
- **Quantitative result:** the model is credible on several 2023 margins, but weak on housing quantities and the historical price path.
- **Central limitation:** the fitted housing response is too weak to restore replacement, so the closed path has no positive stationary endpoint.

Decision: present the finite closed benchmark plus an explicit open sensitivity, or discipline the housing-cost-to-fertility slope and re-estimate under a closed-root requirement before making a between-steady-states claim.

No policy result is promoted until that equilibrium choice is settled.